Project: Worked examples
Title: U-values and room heat loss
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

U-values of built-up elements from their layers and surface resistances, and the steady-state design heat loss of one room: fabric loss through each element to its adjacent temperature, a thermal-bridge allowance, ventilation loss (through the outside temperature or an MVHR supply temperature), the room load for sizing emitters, the heat loss coefficient and the loss per floor area; optionally checked against target U-values.
Element U-value: U = 1/(R_si + Σ d/λ + R_g + R_se), or the user's U override for ground floors and glazing Element fabric loss: Φ_i = A·U·f·(θ_int − θ_adj), θ_adj = θ_e where the element faces outside Thermal bridges: Φ_tb = y_tb·Σ(A·f)_ext·(θ_int − θ_e) on the elements facing outside Fabric total: Φ_T = ΣΦ_i + Φ_tb Ventilation flow: q_v = n·V Ventilation loss: Φ_V = 0.34·q_v·(θ_int − θ_e), or (θ_int − θ_supply) through an MVHR supply, plus 0.34·n_inf·V·(θ_int − θ_e) infiltration with MVHR Room load: Φ_HL = Φ_T + Φ_V (CIBSE DHDG method: × (1 + additional allowance)) Heat loss coefficient: HLC = Φ_HL/(θ_int − θ_e); loss per floor area Φ_HL/A_floor Check: U against the target U of each element

Ref: BS EN ISO 6946 homogeneous layers and surface resistances (clause numbers to be checked against the owner's copy) — Building components — thermal resistance and thermal transmittance
Ref: BS EN 12831-1 cl. 6.3.2 (clause number to be checked against the owner's copy) — Heating systems in buildings — design heat load: transmission heat loss
Ref: BS EN 12831-1 cl. 6.3.3 (clause number to be checked against the owner's copy) — Heating systems in buildings — design heat load: ventilation heat loss
Ref: CIBSE Domestic Heating Design Guide 2026 ed., reduced method (name and edition unverified; to be checked against the owner's copy) — Room heat loss and the additional allowance

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Calculation method (changes the citations; the DHDG option adds the allowance below)	BS EN 12831-1 elemental		assumption: Elemental method; the design-team default.	any
theta_int	Internal design temperature	21	degC	spec: SPEC-A-02 §3.1: living-room design temperature	16 … 24 degC
theta_e	External design temperature (location record)	-3	degC	assumption: Illustrative location design temperature; take the project's own from its design-temperature record.	-15 … 5 degC
layers	Construction layers (several elements share the table, keyed by element name)	8 row(s)		drawing: A-210 rev C	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers[1]		element = Cavity wall; name = Outer brick leaf; d = 102.5 mm; lambda = 0.77 W/(m*K)			
layers[2]		element = Cavity wall; name = PIR insulation; d = 100 mm; lambda = 0.022 W/(m*K)			
layers[3]		element = Cavity wall; name = Inner block leaf; d = 100 mm; lambda = 0.15 W/(m*K)			
layers[4]		element = Cavity wall; name = Plaster; d = 13 mm; lambda = 0.57 W/(m*K)			
layers[5]		element = Party wall; name = Plaster; d = 13 mm; lambda = 0.57 W/(m*K)			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers[6]		element = Party wall; name = Block leaf 1; d = 100 mm; lambda = 0.51 W/(m*K)			
layers[7]		element = Party wall; name = Block leaf 2; d = 100 mm; lambda = 0.51 W/(m*K)			
layers[8]		element = Party wall; name = Plaster; d = 13 mm; lambda = 0.57 W/(m*K)			
elements	Building elements bounding the room	5 row(s)		drawing: A-101 rev C	any
elements[1]		name = Front wall; construction = Cavity wall; area = 8.7 m^2			
elements[2]		name = Window; area = 2.1 m^2; U_user = 1.4 W/(m^2*K)			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
elements[3]		name = Gable wall; construction = Cavity wall; area = 8.64 m^2			
elements[4]		name = Party wall; area = 8.64 m^2; theta_adj = 18 degC; Rg = 0.18 m^2*K/W; U_max = 1.2 W/(m^2*K)			
elements[5]		name = Ground floor; area = 16.2 m^2; U_user = 0.25 W/(m^2*K)			
y_tb	Thermal-bridge allowance on the area facing outside	0.1	W/(m^2*K)	assumption: Illustrative thermal-bridge allowance; replace with the project's bridging record.	0 … 0.3 W/(m^2*K)
U_target	Target U for built-up elements without their own target	0.3	W/(m^2*K)	spec: SPEC-A-02 §4.2: external wall U-value target	0.05 … 8 W/(m^2*K)
n	Air change rate	0.8	1/h	assumption: Natural ventilation allowance for a living room.	0.1 … 3 1/h
V	Room volume	38.88	m^3	drawing: A-101 rev C	5 … 1000 m^3
A_floor	Floor area, for the loss per floor area	16.2	m^2	drawing: A-101 rev C	1 … 200 m^2
f_add	Additional allowance on the room load (CIBSE DHDG method only)	0	%	default	0 … 20 %

3. Assumptions

A1. Steady state: every temperature is constant, with no solar or internal gains, no thermal mass and no

dynamic simulation. (Design heat loss, not an annual energy estimate.)
A2. Layers are homogeneous and in series (one-dimensional flow); no bridged layers (timber studs, mortar joints, fixings) are corrected for: a thermal-bridge allowance y_tb is added instead. (BS EN ISO 6946 homogeneous-layer calculation.)
A3. A blank Rsi is the conventional horizontal-flow 0.13 m²K/W and a blank Rse is 0.04 m²K/W for an element facing outside (no adjacent temperature, or one at or below θ_e), or 0.13 m²K/W where it faces another space; floors, roofs and other flow directions take the user's own values. (Small conventional BS EN ISO 6946 set, not a table of design data.)
A4. Ground-floor U-values and glazing U-values are entered by the user (override); the calculator does not model ground heat transfer. (Stated limit.)
A5. The air constant 0.34 Wh/(m³·K) is the conventional rounded ρ·c_p of air (about 1.2 kg/m³ × 1005 J/(kg·K) ÷ 3600); it is derived, not read from a table. (Derived.)
A6. Design temperatures, conductivities, surface resistances and the thermal-bridge allowance are user records from the user's own licensed copies or manufacturers' data; none is embedded. (§A5.)
A7. method (Calculation method (changes the citations; the DHDG option adds the allowance below)) is taken as BS EN 12831-1 elemental. (Elemental method; the design-team default.)
A8. theta_e (External design temperature (location record)) is taken as -3 degC. (Illustrative location design temperature; take the project's own from its design-temperature record.)
A9. y_tb (Thermal-bridge allowance on the area facing outside) is taken as 0.1 W/(m^2*K). (Illustrative thermal-bridge allowance; replace with the project's bridging record.)
A10. n (Air change rate) is taken as 0.8 1/h. (Natural ventilation allowance for a living room.)
A11. Conventional BS EN ISO 6946 horizontal-flow surface resistances (R_si 0.13, R_se 0.04 outside or 0.13 internal, m²K/W) are used where blank, for: Front wall, Gable wall, Party wall. (Small conventional set; enter your own for floors, roofs and other flow directions.)

4. Working

[8] R_1 = R_si + ☒ d/☒ + R_g + R_se — BS EN ISO 6946
= Front wall: 0.13 + (0.13312 + 4.5455 + 0.66667 + 0.022807) + 0 + 0.04
= 5.538 m²·K/W
Front wall: total thermal resistance

[9] U_1 = 1/R — BS EN ISO 6946
= 1 / 5.538
= 0.18057 W/(m²·K)
Front wall: U-value

[10] Phi_1 = A·U·f·(☒_int − ☒_adj) — BS EN 12831-1 cl. 6.3.2
= 8.7 × 0.18057 × 1 × (21 − -3) °C
= 37.703 W
Front wall: fabric loss to θ_e = -3 °C

[11] U_2 = U override (user-entered)
= Window: 1.4 W/m²K
= 1.4 W/(m²·K)
Window: U-value as entered (ground floor or glazing)

[12] Phi_2 = A·U·f·(☒_int − ☒_adj) — BS EN 12831-1 cl. 6.3.2
= 2.1 × 1.4 × 1 × (21 − -3) °C
= 70.56 W
Window: fabric loss to θ_e = -3 °C

[13] R_3 = R_si + ☒ d/☒ + R_g + R_se — BS EN ISO 6946
= Gable wall: 0.13 + (0.13312 + 4.5455 + 0.66667 + 0.022807) + 0 + 0.04
= 5.538 m²·K/W
Gable wall: total thermal resistance

[14] U_3 = 1/R — BS EN ISO 6946
= 1 / 5.538
= 0.18057 W/(m²·K)

Gable wall: U-value

[15] Phi_3 = A·U·f·(□_int − □_adj) BS EN 12831-1 cl. 6.3.2

= 8.64 × 0.18057 × 1 × (21 − -3) °C

= 37.443 W

Gable wall: fabric loss to θ_e = -3 °C

[16] R_4 = R_si + □ d/□ + R_g + R_se BS EN ISO 6946

= Party wall: 0.13 + (0.022807 + 0.19608 + 0.19608 + 0.022807) + 0.18 + 0.13

= 0.87777 m²·K/W

Party wall: total thermal resistance

[17] U_4 = 1/R BS EN ISO 6946

= 1 / 0.87777

= 1.1392 W/(m²·K)

Party wall: U-value

[18] Phi_4 = A·U·f·(□_int − □_adj) BS EN 12831-1 cl. 6.3.2

= 8.64 × 1.1392 × 1 × (21 − 18) °C

= 29.529 W

Party wall: fabric loss to θ_adj = 18 °C

[19] U_5 = U override (user-entered)

= Ground floor: 0.25 W/m²K

= 0.25 W/(m²·K)

Ground floor: U-value as entered (ground floor or glazing)

[20] Phi_5 = A·U·f·(□_int − □_adj) BS EN 12831-1 cl. 6.3.2

= 16.2 × 0.25 × 1 × (21 − -3) °C

= 97.2 W

Ground floor: fabric loss to θ_e = -3 °C

[21] A_fab = □ A

= 44.28 m²

= 44.28 m²

Total fabric area

[22] A_ext = □ A facing outside

= 35.64 m²

= 35.64 m²

Fabric area facing outside

[23] A_tb = □ A·f facing outside BS EN 12831-1 cl. 6.3.2

= 35.64 m²

= 35.64 m²

Fabric area facing outside weighted by the temperature factor (thermal-bridge allowance applies)

[24] U_m = □ A·U·f / □ A

= 19.964 / 44.28

= 0.45086 W/(m²·K)

Area-weighted mean U-value of the fabric

[25] dT = theta_int - theta_e

= 21 degC - (-3 degC)

= 24 K

Design temperature difference, inside to outside

[26] Phi_el = □ □_i BS EN 12831-1 cl. 6.3.2

= 272.43 W

= 272.43 W

Fabric loss through the elements

[27] Phi_tb = y_tb * A_tb * dT BS EN 12831-1 cl. 6.3.2

= 0.1 W/(m²·K) × 35.64 m² × 24 K

= 85.536 W

Thermal-bridge allowance

[28] Phi_T = Phi_el + Phi_tb BS EN 12831-1 cl. 6.3.2
= 272.43 W + 85.536 W
= 357.97 W
Fabric heat loss
[29] q_v = n * V
= 0.8 1/h × 38.88 m³
= 31.104 m³/h
Ventilation air flow
[30] dT_v = dT
= 24 K
= 24 K
Ventilation temperature difference, inside to outside
[31] Phi_V = 0.34 Wh/(m^3*K) * q_v * dT_v BS EN 12831-1 cl. 6.3.3
= 0.34 Wh/(m^3*K) × 31.104 m³/h × 24 K
= 253.81 W
Ventilation heat loss (0.34 Wh/m³K = conventional ρ·c_p of air, derived)
[32] Phi_HL = Phi_T + Phi_V BS EN 12831-1 cl. 6.3.2
= 357.97 W + 253.81 W
= 611.78 W
Room design heat loss
[33] HLC = Phi_HL / dT
= 611.78 W / 24 K
= 25.491 W/K
Heat loss coefficient
[35] Phi_A = Phi_HL / A_floor
= 611.78 W / 16.2 m²
= 37.764 W/m²
Room design heat loss per floor area
[36] u_1 = U / U_target
= 0.18057 / 0.3
= 0.6019
Front wall: utilisation of the target U-value
[37] u_3 = U / U_target
= 0.18057 / 0.3
= 0.6019
Gable wall: utilisation of the target U-value
[38] u_4 = U / U_target
= 1.1392 / 1.2
= 0.94937
Party wall: utilisation of the target U-value

5. Results

SYMBOL	VALUE	DESCRIPTION
R_1	5.538 m²·K/W	Front wall: thermal resistance
U_1	0.18057 W/(m²·K)	Front wall: U-value
Phi_1	37.703 W	Front wall: fabric loss
U_2	1.4 W/(m²·K)	Window: U-value
Phi_2	70.56 W	Window: fabric loss
R_3	5.538 m²·K/W	Gable wall: thermal resistance
U_3	0.18057 W/(m²·K)	Gable wall: U-value

SYMBOL	VALUE	DESCRIPTION
Phi_3	37.443 W	Gable wall: fabric loss
R_4	0.87777 m²·K/W	Party wall: thermal resistance
U_4	1.1392 W/(m²·K)	Party wall: U-value
Phi_4	29.529 W	Party wall: fabric loss
U_5	0.25 W/(m²·K)	Ground floor: U-value
Phi_5	97.2 W	Ground floor: fabric loss
A_fab	44.28 m²	Total fabric area
A_ext	35.64 m²	Fabric area facing outside
A_tb	35.64 m²	Fabric area facing outside weighted by f
U_m	0.45086 W/(m²·K)	Mean fabric U-value
dT	24 K	Design temperature difference
Phi_el	272.43 W	Element fabric loss
Phi_tb	85.536 W	Thermal-bridge allowance
Phi_T	357.97 W	Fabric heat loss
q_v	31.104 m³/h	Ventilation air flow
Phi_V	253.81 W	Ventilation heat loss
Phi_HL	611.78 W	Room design heat loss
HLC	25.491 W/K	Heat loss coefficient
Phi_A	37.764 W/m²	Heat loss per floor area
u_1	0.6019	Front wall: U against target
u_3	0.6019	Gable wall: U against target
u_4	0.94937	Party wall: U against target

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	U-value of Party wall against its target (1.2 W/m²K)	u_4 = 0.94937	u_4 ≤ 1	0.949	
PASS	U-value of Front wall against its target (0.3 W/m²K)	u_1 = 0.6019	u_1 ≤ 1	0.602	
PASS	U-value of Gable wall against its target (0.3 W/m²K)	u_3 = 0.6019	u_3 ≤ 1	0.602	

7. Applicability

WARNING — record.theta_e: theta_e: a tabulated value not taken from a reference record (External design temperature); enter it as one so its source, edition and licence travel with it.
WARNING — record.elements: elements: a tabulated value not taken from a reference record (Surface resistances and air-gap resistance); enter it as one so its source, edition and licence travel with it.
WARNING — record.layers: layers: a tabulated value not taken from a reference record (Layer thermal

conductivity); enter it as one so its source, edition and licence travel with it.
WARNING — record.y_tb: y_tb: a tabulated value not taken from a reference record (Thermal-bridge allowance); enter it as one so its source, edition and licence travel with it.
WARNING — input.f_add: f_add (Additional allowance on the room load (CIBSE DHDG method only)) was not entered; the template default 0 % is used.

8. Verification

```
Template bs.room-heat-loss@1.0.0 Content hash c3e6168e43239acd Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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